

for example, that when  $\Delta = \Delta^{RM}(\bar{M})$ , the power of the conditional test converges to the power envelope when there is a unique (non-zero) pre-treatment maximum violation, *i.e.* when  $\max_{s < 0} |\delta_{s+1} - \delta_s| > 0$  and has a unique solution.<sup>24</sup> Likewise, the conditional test has optimal local asymptotic power for  $\Delta^{SDRM}(\bar{M})$  when there is a unique maximum non-linearity in the pre-treatment period. Intuitively, this is because the upper bound of the identified set is determined by a single  $\Delta_{k^*}$  satisfying LICQ, so the conditional test for this  $\Delta_{k^*}$  has optimal local asymptotic power, whereas our consistency results imply that the tests for the remaining  $\Delta_k$  that do not determine the identified set bound reject with probability approaching 1. See Corollary 4.1 in the working paper version of this paper for a formal derivation (Rambachan and Roth, 2021).

Proposition 3.3 shows that under LICQ the local asymptotic power of the conditional test converges to the power envelope for tests controlling size in the finite-sample normal model. In the working paper version of this paper, we showed that the power envelope from the finite-sample normal model corresponds with the power-envelope among tests that control size asymptotically and have certain invariance properties using results in Müller (2011) (Proposition E.4 in Rambachan and Roth, 2021).

#### 4. INFERENCE USING FIXED LENGTH CONFIDENCE INTERVALS

We next consider FLCIs based on affine estimators. While the conditional and hybrid confidence sets offer attractive asymptotic power guarantees under asymptotics in which sampling variation grows small relative to the length of the identified set, FLCIs offer finite-sample power guarantees (in the normal model) for certain classes  $\Delta$  of interest. In certain special cases, FLCIs may thus outperform the ARP tests when sampling variation is large relative to the length of the identified set. For brevity of exposition, we focus on the properties of FLCIs in the case where the finite-sample normal approximation (9) holds exactly with  $\Sigma_n$  known; Armstrong and Kolesár (2020b) provide uniform asymptotic results for FLCIs under conditions similar to those in Section 3.3.

##### 4.1. Constructing FLCIs

Following Donoho (1994) and Armstrong and Kolesár (2018, 2020a), we consider FLCIs based on an affine estimator for  $\theta$ , denoted by  $\mathcal{C}_{a,n}(a, v, \chi) := (a + v'\hat{\beta}_n) \pm \chi$ , where  $a$  and  $\chi$  are scalars and  $v \in \mathbb{R}^{T+\bar{T}}$ . We minimize the half-length of the confidence interval,  $\chi$ , subject to the constraint that  $\mathcal{C}_{a,n}(a, v, \chi)$  satisfies the coverage requirement (10) in the finite-sample normal model.

Observe that if  $\hat{\beta}_n \sim \mathcal{N}(\beta, \Sigma_n)$ , then  $a + v'\hat{\beta}_n \sim \mathcal{N}(a + v'\beta, v'\Sigma_n v)$ , and hence  $|a + v'\hat{\beta}_n - \theta| \sim |\mathcal{N}(b, v'\Sigma_n v)|$ , where  $b = a + v'\beta - \theta$  is the affine estimator's bias for  $\theta$ . Observe further that  $\theta \in \mathcal{C}_{a,n}(a, v, \chi)$  if and only if  $|a + v'\hat{\beta}_n - \theta| \leq \chi$ . For fixed values  $a$  and  $v$ , the smallest value of  $\chi$  that satisfies (10) is therefore the  $1 - \alpha$  quantile of the  $|\mathcal{N}(\bar{b}, v'\Sigma_n v)|$  distribution, where  $\bar{b}$  is the affine estimator's worst-case bias

$$\bar{b}(a, v) := \sup_{\delta \in \Delta, \tau_{post} \in \mathbb{R}^{\bar{T}}} |a + v'(\delta + L_{post}\tau_{post}) - l'\tau_{post}|. \quad (17)$$

24. For this convergence to hold uniformly, the non-binding moments must be slack by  $\epsilon$ , so we would need that  $\max_{s < 0} |\delta_{s+1} - \delta_s|$  is at least  $\epsilon$  greater than the second largest difference.

Let  $cv_\alpha(t)$  denote the  $1 - \alpha$  quantile of the folded normal distribution  $|\mathcal{N}(t, 1)|$ .<sup>25</sup> For fixed  $a$  and  $v$ , the smallest value of  $\chi$  satisfying the coverage requirement (10) is thus

$$\chi_n(a, v; \alpha) = \sigma_{v,n} \cdot cv_\alpha(\bar{b}(a, v)/\sigma_{v,n}), \quad (18)$$

where  $\sigma_{v,n} := \sqrt{v' \Sigma_n v}$ . The optimal (*i.e.* minimum-length) FLCI is constructed by choosing the values of  $a$  and  $v$  to minimize (18). When  $\Delta$  is convex, this minimization can be solved as a nested optimization problem, where both the inner and outer minimizations are convex (Low, 1995; Armstrong and Kolesár, 2018, 2020a). We denote the  $1 - \alpha$  level, optimal FLCI by  $\mathcal{C}_{\alpha,n}^{FLCI}(\hat{\beta}_n, \Sigma_n) := (a_n + v_n' \hat{\beta}_n) \pm \chi_n$ , where  $\chi_n := \inf_{a,v} \chi_n(a, v; \alpha)$  and  $a_n, v_n$  are the optimal values in the minimization.

**4.1.1. Example:  $\Delta^{\text{SD}}(M)$ .** Suppose  $\theta = \tau_1$ . For  $\Delta^{\text{SD}}(M)$ , the affine estimator used by the optimal FLCI takes the form  $a + v' \hat{\beta}_n = \hat{\beta}_{n,1} - \sum_{s=-T+1}^0 w_s (\hat{\beta}_{n,s} - \hat{\beta}_{n,s-1})$ , where the weights  $w_s$  sum to one (but may be negative). This estimator adjusts the event-study coefficient for  $t = 1$  by an estimate of the differential trend between  $t = 0$  and  $t = 1$  formed by taking a weighted average of the differential trends in periods prior to treatment. The worst-case bias will be smaller if more weight is placed on pre-treatment periods closer to the treatment date, but it may reduce variance to place more weight on earlier pre-periods. The weights  $w_s$  are optimally chosen to balance this trade-off.

#### 4.2. Finite-sample near optimality

In particular cases of interest, such as when  $\Delta = \Delta^{\text{SD}}(M)$ , the optimal FLCIs introduced above have near-optimal expected length in the finite-sample normal model. The following result, which is an immediate consequence of results in Armstrong and Kolesár (2018, 2020a), bounds the ratio of the expected length of the shortest possible confidence interval that controls size relative to the length of the optimal FLCI.

**Assumption 8.** Assume (i)  $\Delta$  is convex and centrosymmetric (*i.e.*  $\tilde{\delta} \in \Delta$  implies  $-\tilde{\delta} \in \Delta$ ), and (ii)  $\delta \in \Delta$  is such that  $(\tilde{\delta} - \delta) \in \Delta$  for all  $\tilde{\delta} \in \Delta$ .

**Proposition 4.1.** Suppose  $\delta$  and  $\Delta$  satisfy Assumption 8. Let  $\mathcal{I}_\alpha(\Delta, \Sigma_n)$  denote the class of confidence sets that satisfy the coverage criterion (10) at the  $1 - \alpha$  level. Then, for any  $\tau$  with  $\tau_{\text{pre}} = 0$  and  $\Sigma_n$  positive definite,

$$\frac{\inf_{C_{\alpha,n} \in \mathcal{I}_\alpha(\Delta, \Sigma_n)} \mathbb{E}_{\hat{\beta}_n \sim \mathcal{N}(\delta + \tau, \Sigma_n)} [\lambda(C_{\alpha,n})]}{2\chi_n} \geq \frac{z_{1-\alpha}(1 - \alpha) - \tilde{z}_\alpha \Phi(\tilde{z}_\alpha) + \phi(z_{1-\alpha}) - \phi(\tilde{z}_\alpha)}{z_{1-\alpha/2}},$$

where  $\lambda(\cdot)$  denotes the length (Lebesgue measure) of a set and  $\tilde{z}_\alpha = z_{1-\alpha} - z_{1-\alpha/2}$ .

Part (i) of Assumption 8 is satisfied for  $\Delta^{\text{SD}}(M)$  but not for our other ongoing examples. For example,  $\Delta^{\text{SDPB}}(M)$  is convex but not centrosymmetric, and  $\Delta^{\text{RM}}(\bar{M})$  is neither convex nor centrosymmetric. Part (ii) of Assumption 8 is satisfied whenever parallel trends holds in both the pre-treatment and post-treatment periods ( $\delta = 0$ ) and whenever  $\delta$  is a linear trend for the case of  $\Delta^{\text{SD}}(M)$ .

FLCIs thus offer attractive guarantees for the case of  $\Delta^{\text{SD}}(M)$ . When  $\alpha = 0.05$ , the lower bound in Proposition 4.1 evaluates to 0.72, meaning that the expected length of the shortest

25. If  $t = \infty$ , we define  $cv_\alpha = \infty$ .

possible confidence set that satisfies the coverage requirement (10) is at most 28% shorter than the length of the optimal FLCI when the conditions of the proposition hold.

#### 4.3. (In)Consistency of FLCIs

As discussed above, these finite-sample guarantees do not apply for several types of restrictions  $\Delta$  of importance, including those that construct bounds using the maximum pre-treatment violation or incorporate sign and shape restrictions. We now show that the FLCIs can perform poorly under such restrictions. We first provide two illustrative examples, and then state a formal inconsistency result.

**4.3.1. Example:  $\Delta^{SDPB}(M)$  and  $\Delta^{SDI}(M)$ .** Suppose  $\theta = \tau_1$ . It can be shown that the worst-case bias of an affine estimator over  $\Delta^{SDPB}(M)$  or  $\Delta^{SDI}(M)$  is the same as the worst-case bias for that estimator over  $\Delta^{SD}(M)$ .<sup>26</sup> Since the construction of the optimal FLCI depends only on the worst-case bias and variance of the affine estimator, it follows that the optimal FLCI constructed using  $\Delta^{SDPB}(M)$  or  $\Delta^{SDI}(M)$  is the same as the one constructed using  $\Delta^{SD}(M)$ . Therefore, the optimal FLCI does not adapt to additional sign or monotonicity restrictions.

**4.3.2. Example:  $\Delta^{RM}(\bar{M})$ .** Suppose  $\theta = \tau_1$ . If  $\Delta = \Delta^{RM}(\bar{M})$  and  $\bar{M} > 0$ , then all affine estimators for  $\tau_1$  have infinite worst-case bias, since  $\delta \in \Delta^{RM}(\bar{M})$  can have  $|\delta_1|$  arbitrarily large if  $|\delta_{-1}|$  is also sufficiently large. Thus, the only valid FLCI is the entire real line.

We next provide a formal result on the (in)consistency of the FLCIs. Specifically, we will show that even as the sampling variation  $\Sigma_n$  converges to 0, the optimal FLCI will include fixed points outside of the identified set with positive probability unless certain special conditions are met.<sup>27</sup> Recall from Lemma 2.1 that the identified set  $S(\beta, \Delta)$  is an interval when  $\Delta$  is convex, with length equal to  $\theta^{ub}(\beta, \Delta) - \theta^{lb}(\beta, \Delta) = b^{max}(\beta_{pre}, \Delta) - b^{min}(\beta_{pre}, \Delta)$ . Since the length of the identified set only depends on  $\beta_{pre}$  and  $\Delta$ , denote it by  $LID(\beta_{pre}, \Delta)$ . Our next result shows that  $C_{\alpha,n}^{FLCI}(\hat{\beta}_n, \Sigma_n)$  is consistent if and only if  $LID(\beta_{pre}, \Delta)$  is its maximum possible value, provided that the identified set is not the entire real line (in which case any procedure is trivially consistent).

**Assumption 9** (Identified set maximal length and finite). *Suppose  $\delta \in \Delta$  is such that  $LID(\delta_{pre}, \Delta) = \sup_{\tilde{\delta}_{pre} \in \Delta_{pre}} LID(\tilde{\delta}_{pre}, \Delta) < \infty$ , where  $\Delta_{pre} = \{\delta_{pre} \in \mathbb{R}^T : \exists \delta_{post} \text{ s.t. } (\delta'_{pre}, \delta'_{post})' \in \Delta\}$  is the set of possible values for  $\delta_{pre}$ .*

**Proposition 4.2.** *Suppose  $\Delta$  is convex and  $\alpha < 0.5$ . Fix  $\delta \in \Delta$  and  $\tau$  with  $\tau_{pre} = 0$ , and suppose  $S(\delta + \tau, \Delta) \neq \mathbb{R}$ . Then  $(\delta, \Delta)$  satisfy Assumption 9 if and only if  $C_{\alpha,n}^{FLCI}(\hat{\beta}_n, \Sigma_n)$  is consistent, meaning that for  $\Sigma_n = \Sigma^*/n$ ,*

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\hat{\beta}_n \sim \mathcal{N}(\delta + \tau, \Sigma_n)} \left( \theta^{out} \in C_{\alpha,n}^{FLCI}(\hat{\beta}_n, \Sigma_n) \right) = 0 \quad \text{for all } \theta^{out} \notin S(\delta + \tau, \Delta).$$

26. Suppose the vector  $\tilde{\delta}$  maximizes the bias for an affine estimator  $(a, v)$  over  $\Delta^{SD}(M)$ . The vector that adds a constant slope to  $\tilde{\delta}$ , say  $\tilde{\delta}_c = \tilde{\delta} + c \cdot (-T, \dots, T)'$ , also lies in  $\Delta^{SD}(M)$ , and for  $c$  sufficiently large,  $\tilde{\delta}_c$  will lie in  $\Delta^{SDPB}(M)$ . Moreover, the worse-case bias will be the same for  $\tilde{\delta}$  and  $\tilde{\delta}_c$ , since if  $(a, v)$  has finite worst-case bias it must subtract out a weighted average of the pre-treatment slopes.

27. For ease of exposition, we present a result using “small- $\Sigma$ ” asymptotics in the normal model, as in e.g. Kadane (1971) and Moreira and Ridder (2019).

Thus, if Assumption 9 fails, then  $\mathcal{C}_{a,n}^{FLCI}(\hat{\beta}_n, \Sigma_n)$  is inconsistent in the strong sense that it includes fixed points outside of the identified set with non-vanishing probability. It follows that there will be some  $\delta \in \Delta$  such that the FLCI is inconsistent under  $\delta$  unless the identified set is always the same length. Proposition 4.2 is new, and may be relevant for other settings in which FLCIs are used.

The intuition for the possible inconsistency of FLCIs is as follows: to ensure that an FLCI satisfies the coverage requirement (10), its length must be at least  $\sup_{\delta_{pre} \in \Delta_{pre}} LID(\delta_{pre}, \Delta)$ . However, this implies that if in fact  $LID(\delta_{pre}, \Delta) < \sup_{\delta_{pre} \in \Delta_{pre}} LID(\delta_{pre}, \Delta)$ , then the FLCI is strictly longer than the length of the identified set, regardless of the value of  $\Sigma_n$ , and thus some points outside of the identified set must be covered with non-vanishing probability. This reflects the fact that FLCIs are by construction *fixed length*, and thus their length does not adapt to information in the data about the length of the identified set. By contrast, the length of the conditional/hybrid confidence sets can depend on  $\hat{\beta}_{pre}$  and thus “adapts” to the length of the identified set.

In the three-period DiD example, Assumption 9 holds everywhere for  $\Delta^{SD}(M)$  (since the identified set is always the same length,  $2M$ ), for values of  $\delta$  where the sign restrictions do not bind for  $\Delta^{SDPB}(M)$ , and nowhere for the polyhedra that form  $\Delta^{RM}(\bar{M})$ . The restrictiveness of Assumption 9 thus depends greatly on  $\Delta$ .

The results in this section establish that when certain conditions on  $\Delta$  are satisfied, optimal FLCIs are consistent and have desirable finite-sample guarantees in terms of expected length. FLCIs are thus attractive for our baseline smoothness class  $\Delta^{SD}(M)$ , since they are guaranteed to be consistent and offer attractive finite-sample guarantees. Our inconsistency result shows, however, that FLCIs may perform poorly for other choices of  $\Delta$  that may be of interest in empirical applications, such as those that construct bounds using a pre-treatment maximum or incorporate sign and monotonicity restrictions.

## 5. SIMULATION STUDY

In this section, we conduct a simulation study to investigate the performance of the discussed confidence sets across a range of relevant data-generating processes. We find good size control for all of the procedures, and therefore focus in the main text on a comparison of power to provide concrete recommendations on the best approach in practice. In the [supplementary material](#), we present results on size control and other additional simulation results.

### 5.1. Simulation design

Our simulations are calibrated using the estimated covariance matrix from the 12 recently-published papers surveyed in Roth (2022). For any given paper in the survey, we denote by  $\hat{\Sigma}$  the estimated variance-covariance matrix from the event-study in the paper, calculated using the clustering scheme specified by the authors. For a chosen mean vector  $\beta$ , we simulate event-study coefficients  $\hat{\beta}_s$  from a normal model,  $\hat{\beta}_s \sim \mathcal{N}(\beta, \hat{\Sigma})$ .<sup>28</sup> In simulation  $s$ , we construct nominal 95% confidence sets for the parameter of interest  $\theta$  using the pair  $(\hat{\beta}_s, \hat{\Sigma})$  for each proposed procedure. The parameter of interest is the causal effect in the first post-treatment period ( $\theta = \tau_1$ ); in the [supplementary material](#), we present simulation results in which the parameter of interest

28. We focus on the normal simulations in the main text since it allows for a tractable computation of the optimal excess length of procedures that control size. In the [supplementary material](#), we show that our procedures perform similarly in simulations based on the empirical distribution in the original paper.